

Co-Designing MEMoPAD: Multimodal Emotion Monitoring in clinical Pathways for Anxiety Disorders

Luigi A. Moretti 1, Carmel McGrath 2,3,4, Lucy Condon 3, Miles Thompson 5, Paul Matthews 6, Michael Loizou 7, and David Western 1

1 UWE Bristol (University of the West of England), School of Engineering, U.K.

2 NIHR Health Protection Research Unit in Evaluation and Behavioural Science, Population Health Sciences, Bristol Medical School, University of Bristol, U.K.

3 The National Institute for Health and Care Research Applied Research Collaboration West University Hospitals Bristol and Weston NHS Foundation Trust, U.K.

4 UWE Bristol (University of the West of England), Faculty of Health and Applied Sciences, School of Health and Social Wellbeing, U.K.

5 UWE Bristol (University of the West of England), School of Social Sciences, U.K.

6 UWE Bristol (University of the West of England), School of Computing and Creative Technologies, U.K.

7 University of Plymouth, Faculty of Health, England, U.K.

Keywords: mhealth, affective computing, multimodal, Ecological Momentary Assessment (EMA), anxiety disorders, clinical implementation, emotion recognition, co-design, PPIE

Introduction

Anxiety disorders consistently present challenges of under-/mis-diagnosis due to difficulties in differential diagnoses and the presence of common comorbidities. Standard self-reported approaches for assessing emotional regulation face criticism for their paradoxical nature, while semi-structured interviews may lack specificity and cross-cultural validity [1]. While digital mental health tools are common, their integration into clinical pathways is rare, hampered by high user dropout. To address this, we used a multi-stage co-design process to develop a tool that identifies emotion patterns from bio-signals collected from wearables, along with user-generated data analysed for tone and content. Information from both mobilephones and wearables also allow to contextualise both the emotion data collected and the suggested coping strategies. Acknowledging the historical challenge of dropouts in digital tools for mental health [1], our study embraces a multi-stage co-design approach [2]. This case study summarises initial findings from the co-design activities completed so far.

Methodology

Initial public consultations with people with lived experience of anxiety (hereafter 'patients', n=8) and clinicians (n=6), analysis of studies from the literature, training and guiding materials provided by Health Innovation West of England, and advices from experts of the field –including some co-authors of this work- shaped our approach to co-design; which involves patients, unpaid carers, and clinicians across six phases [3]. Three phases are complete (see Table 1). Each phase includes 2h patient focus groups (up to six participants each to encourage group discussion and exchange of views) and separate 30min one-to-one interviews with carers and clinicians to mitigate potential power imbalances. All co-design activities have been conducted on-line following the majority of participants' preference, updates have been shared via a monthly newsletter and a dedicated website [4] to keep participants and stakeholders informed and updated about the project work. Data collected using PPIE tools (see Table 1) and recordings from Microsoft Teams was analysed by the first author using thematic analysis (Phases I and II) or affinity diagrams (Phase III), followed by comparative analysis for different stakeholder groups' correlations, and a 1h roundtable with interdisciplinary experts, included as co-authors in this work. Each co-design phase started with a presentation of the findings from the previous phase to ensure that all the participants from the different stakeholder groups were informed.

Key findings

Our co-design approach allowed us to build on our existing work [5] and to date, we have achieved the following: three main opportunities for intervention ('scenarios') were prioritised in Phase I (n=24): (a) clinical conversations, (b) anxiety loops, and (c) early help-seeking behaviour. In Phase II (n=20), it was determined that a mobile app linked to a smartwatch could support data collection. This may enable the collection of structured, longitudinal and interpretable data across all three scenarios. To maximise effectiveness and engagement, it is fundamental to personalise the user experience (UX) and contextualise longitudinal data with respect to subjective experiences and daily activities. Ideas and inspirations for drafting the UX and user interface of this digital tool have then been explored in Phase III (n=19). We have also explored the idea to include a web app for sharing data with clinicians, the role generative AI should have, and how the UX should practically and emotionally support carers.

Table 1. Breakdown of Co-design Phases I to III

Co-design	Research question	Key PPIE tools	Participants
Phase I, 'Pre-design'	What are the challenges and the unresolved needs that the stakeholders face due to anxiety disorders?	Online survey	15 patients, 3 carers, 6 clinicians
Phase II, 'Generative'	How can affective computing (i.e. emotion recognition technology) be useful to address and mitigate the challenges defined in the 3 scenarios of reference?	Custom storyboard templates	12 patients, 4 carers, 4 clinicians
Phase III, 'Prototyping'	How should this digital tool look and behave in the three different scenarios of reference? What should be the role of the wearable devices?	Custom mock-up templates	15 patients, 2 carers, 2 clinicians

Discussion

The involvement of local communities has allowed to have a diversified pool of participants, including exponents of minorities and different age groups. We learned that synchronous tools (e.g., collaborative whiteboards), which also require a certain level of digital literacy, could be counterproductive for anxiety-related discussions. Offering lower-pressure, asynchronous and analogue alternatives (e.g., sharing photos of mock-ups via WhatsApp) proved more inclusive and effective. Our patient participants strongly advocated for the inclusion of carers. This underscores a critical lesson for mental health tech design: recruitment and design must address the entire care ecosystem, ranging from personal network, and covering local communities, primary (i.e., GPs) and secondary care. However, a constant reduction of clinicians involvement in our co-design activities has been noticed. Our key takeaway is the need to design flexible, 'low-touch' engagement pathways for time-poor experts, such as brief online interviews, rather than longer focus groups, and asynchronous feedback.

Future plans

Co-design Phase IV, planned for July 2025, aims to gather feedback about the demo of the mobile and wearable app for personal use and the web app for clinicians. By winter 2026 the last two phases will focus on testing the working prototype in lab and wild settings.

References

1. Linardon, Jake, et al. Current evidence on the efficacy of mental health smartphone apps for symptoms of depression and anxiety. A meta-analysis of 176 randomized controlled trials. *World Psychiatry* 23.1 (2024): 139-149.
2. M. Berking and P. Wupperman, 'Emotion regulation and mental health: recent findings, current challenges, and future directions', *Curr. Opin. Psychiatry*, vol. 25, no. 2, pp. 128–134, Mar. 2012.
3. Noorbergen, T.J., Adam, M.T., Teubner, T. and Collins, C.E., 2021. Using co-design in mobile health system development: a qualitative study with experts in co-design and mobile health system development. *JMIR mHealth and uHealth*, 9(11).
4. <https://memopad.luigiandreamoretti.com/>
5. Moretti, L., Thompson, M., Matthews, P., Loizou, M. and Western, D., 2025. Affective Computing in Anxiety Disorders: A Rapid Literature Review of Emotion Recognition Applications. In *Proceedings of the 18th International Joint Conference on Biomedical Engineering Systems and Technologies*.